

Complete Setup & Testing Guide

After FastAPI Implementation

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Prerequisites

System Requirements

- **Python:** 3.11 or higher
- **Operating System:** Windows 11, macOS, or Linux
- **RAM:** 4GB minimum (8GB recommended)
- **Disk Space:** 500MB for dependencies

Required Accounts

- **OpenAI API Key** - Get from <https://platform.openai.com/api-keys>
-

Installation

Step 1: Extract the FastAPI Backend

```
# Navigate to your project folder
cd /path/to/your/project

# Extract the zip file (if not already done)
unzip -o FastAPI_Backend_PDF.zip
```

Step 2: Install Dependencies

```
# Install all required packages
pip install -r requirements.txt
```

Expected output:

```
Successfully installed fastapi-0.104.0 uvicorn-0.24.0 python-multipart-0.0.6
...
```

Step 3: Install Playwright (for PDF generation)

```
# Install Playwright browsers
playwright install chromium
```

Note: This is optional but recommended for PDF export functionality.

Configuration

Step 1: Create Environment File

Create a `.env` file in your project root:

```
# Windows (PowerShell)
New-Item .env -ItemType File

# Mac/Linux
touch .env
```

Step 2: Add Your OpenAI API Key

Edit `.env` and add:

```
OPENAI_API_KEY=sk-your-api-key-here
```

Important: Never commit this file to Git!

Step 3: Verify Configuration

```
# Test if environment variables are loaded
python -c "from dotenv import load_dotenv; import os; load_dotenv(); print('API Key:', 'Found' if os.getenv('OPENAI_API_KEY') else 'Not Found')"
```

Expected output:

```
API Key: Found
```

Running the Application

You have **two options**: FastAPI backend (for React frontend) or Gradio UI (for testing).

Option 1: Run FastAPI Backend (Recommended for Production)

```
# Development mode (with auto-reload)
uvicorn api_server:app --reload --host 0.0.0.0 --port 8000
```

Expected output:

```
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:      Started reloader process
INFO:      Started server process
INFO:      Waiting for application startup.
INFO:      Application startup complete.
```

Access Points: - API Base: <http://localhost:8000> - Interactive Docs: <http://localhost:8000/docs> - Alternative Docs: <http://localhost:8000/redoc>

Option 2: Run Gradio UI (For Internal Testing)

```
# Run Gradio interface
python gradio_ui.py
```

Expected output:

```
Running on local URL:  http://127.0.0.1:7860
```

Option 3: Run Both Simultaneously

Terminal 1 (FastAPI):

```
uvicorn api_server:app --reload --port 8000
```

Terminal 2 (Gradio):

```
python gradio_ui.py
```

Now you have: - FastAPI: <http://localhost:8000> (for React frontend) - Gradio: <http://localhost:7860> (for your testing)

Testing the API

Test 1: Health Check

```
curl http://localhost:8000/health
```

Expected response:

```
{
  "status": "healthy",
  "timestamp": "2025-11-24T17:00:00.000000",
  "agents": {
    "tax_agent": "available",
    "bill_agent": "available"
  }
}
```

✓ **Pass:** Server is running correctly

Test 2: Get Default Bill Invoice State

```
curl http://localhost:8000/api/bill/default
```

Expected response:

```
{
  "success": true,
  "state": {
    "freight_bill": {
      "series_no": "",
      "runs": [],
      ...
    }
  },
  "message": "Default bill invoice state"
}
```

✓ **Pass:** API is returning data correctly

Test 3: Process File Upload (Windows PowerShell)

```
# Create a test request
$headers = @{}
$form = @{}
    file = Get-Item -Path "path\to\your\file.xlsx"
    state = '{"freight_bill":{"runs":[]}}'
    command = "add all details from dataset to invoice"
}

Invoke-RestMethod -Uri "http://localhost:8000/api/bill/process" -Method Post -
Form $form
```

Test 3: Process File Upload (Mac/Linux)

```
curl -X POST http://localhost:8000/api/bill/process \
-F "file=@/path/to/your/file.xlsx" \
-F "state={'freight_bill':{'runs':[]}}" \
-F "command=add all details from dataset to invoice"
```

Expected response:

```
{
  "success": true,
  "state": {
    "freight_bill": {
      "runs": [
        {
          "date": "21-10-2025",
          "truck_no": "AP39TU3699",
          ...
        }
      ]
    }
  },
  "message": "✓ Agentic operation complete: ADD"
}
```

✓ **Pass:** File upload and extraction working

Test 4: Interactive Testing (Easiest)

1. Open browser: <http://localhost:8000/docs>
2. Click on any endpoint (e.g., `GET /api/bill/default`)
3. Click "Try it out"
4. Click "Execute"
5. See the response below

This is the EASIEST way to test all endpoints!

Testing with Gradio UI

Step 1: Open Gradio Interface

```
python gradio_ui.py
```

Open: <http://localhost:7860>

Step 2: Test Bill Invoice

1. Click "Bill Invoice" tab
2. Upload your Excel file

3. Enter command: "add all details from dataset to invoice"
4. Click "**Process with Agent**"
5. See the preview update with extracted data

✓ **Pass:** Gradio UI working with agents

Step 3: Test Export

1. After processing, click "**Export Bill PDF**"
2. PDF should download automatically

✓ **Pass:** Export functionality working

Troubleshooting

Issue 1: "Module not found" Error

Problem:

```
ModuleNotFoundError: No module named 'fastapi'
```

Solution:

```
pip install -r requirements.txt
```

Issue 2: "Port already in use"

Problem:

```
ERROR: [Errno 48] Address already in use
```

Solution:

Windows:

```
# Find process using port 8000
netstat -ano | findstr :8000

# Kill the process (replace PID with actual number)
taskkill /PID <PID> /F
```

Mac/Linux:

```
# Find and kill process
lsof -ti:8000 | xargs kill -9
```

Or use a different port:

```
uvicorn api_server:app --port 8001
```

Issue 3: "OpenAI API Key not found"

Problem:

```
Error: OPENAI_API_KEY not found
```

Solution: 1. Check `.env` file exists 2. Verify API key is correct 3. Restart the server after adding key

Issue 4: "0 rows extracted" from Excel

Problem: Agent extracts 0 rows even though file has data.

Solution:  **Already fixed!** The bug fixes in the zip handle this.

If still happening: 1. Check if file has correct sheet (EXY1) 2. Verify columns have data 3. Check `context_bill_invoice.py` has the bug fixes

Issue 5: PDF Export Not Working

Problem:

```
PDF export failed: Playwright not installed
```


Solution:

```
pip install playwright
playwright install chromium
```

Alternative: Use HTML export instead:

```
{
  "state": {...},
  "format": "html"
}
```

Production Deployment

Step 1: Install Production Server

```
pip install gunicorn
```

Step 2: Run with Gunicorn (Linux/Mac)

```
gunicorn api_server:app \
  --workers 4 \
  --worker-class uvicorn.workers.UvicornWorker \
  --bind 0.0.0.0:8000 \
  --access-logfile access.log \
  --error-logfile error.log
```

Step 3: Run as Background Service (Linux)

Create `/etc/systemd/system/invoice-api.service`:

```
[Unit]
Description=Invoice API Service
After=network.target

[Service]
Type=notify
User=your-username
WorkingDirectory=/path/to/invoice_app
Environment="PATH=/path/to/venv/bin"
ExecStart=/path/to/venv/bin/gunicorn api_server:app \
  --workers 4 \
  --worker-class uvicorn.workers.UvicornWorker \
  --bind 0.0.0.0:8000

[Install]
WantedBy=multi-user.target
```

Enable and start:

```
sudo systemctl enable invoice-api
sudo systemctl start invoice-api
sudo systemctl status invoice-api
```

Step 4: Setup Nginx Reverse Proxy

Create `/etc/nginx/sites-available/invoice-api`:

```
server {
    listen 80;
    server_name your-domain.com;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

Enable:

```
sudo ln -s /etc/nginx/sites-available/invoice-api /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx
```

Quick Reference Commands

Start Development Server

```
uvicorn api_server:app --reload --port 8000
```

Start Gradio UI

```
python gradio_ui.py
```

Test Health

```
curl http://localhost:8000/health
```

View Logs

```
# If running in foreground: logs appear in terminal  
# If running in background:  
tail -f api_server.log
```

Stop Server

```
# Press CTRL+C in terminal  
# Or kill process:  
pkill -f "uvicorn api_server"
```

Summary Checklist

Before going live, verify:

-  Python 3.11+ installed
-  All dependencies installed (`pip install -r requirements.txt`)
-  Playwright installed (`playwright install chromium`)
-  `.env` file with OpenAI API key
-  FastAPI server starts without errors

-  Health check returns "healthy"
 -  Can get default state
 -  Can process file upload
 -  Can export to HTML/PDF
 -  Gradio UI works (optional)
-

Need Help?

1. **Check logs:** Look at terminal output for errors
 2. **Test endpoints:** Use `http://localhost:8000/docs`
 3. **Verify files:** Ensure all files from zip are extracted
 4. **Check API key:** Verify OpenAI key is valid
 5. **Review documentation:** Read `API_DOCUMENTATION.pdf`
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You're all set! 

Start the server and begin testing with your React frontend or Gradio UI!